



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI

**MODULE 2 SCALABLE DATA SOLUTIONS:
BIG DATA ENGINEERING ESSENTIALS
ASSIGNMENT PROJECT**

TEAM 3
BALKIS AMMAL
SONG QIU XUAN
MAO JIN PING
MAEGALA
KO LI ZHOU

20 JUNE 2026

Unlocking R\$3.6M in Hidden Revenue

Transforming Olist's Data
Infrastructure into an Engine
for Customer Retention

EXECUTIVE ARCHITECTURE & STRATEGY REVIEW

Transforming Raw Commerce Data into High-Value Customer Retention



The Problem

Customer acquisition costs are rising, yet **75%** of Olist customers purchase only once. We lack visibility into who will return and who will churn.



The Engine

A fully automated, decoupled **ELT pipeline** built on **BigQuery** and **dbt**. Ingests ~550K raw records and refines them into strict, scalable business models.

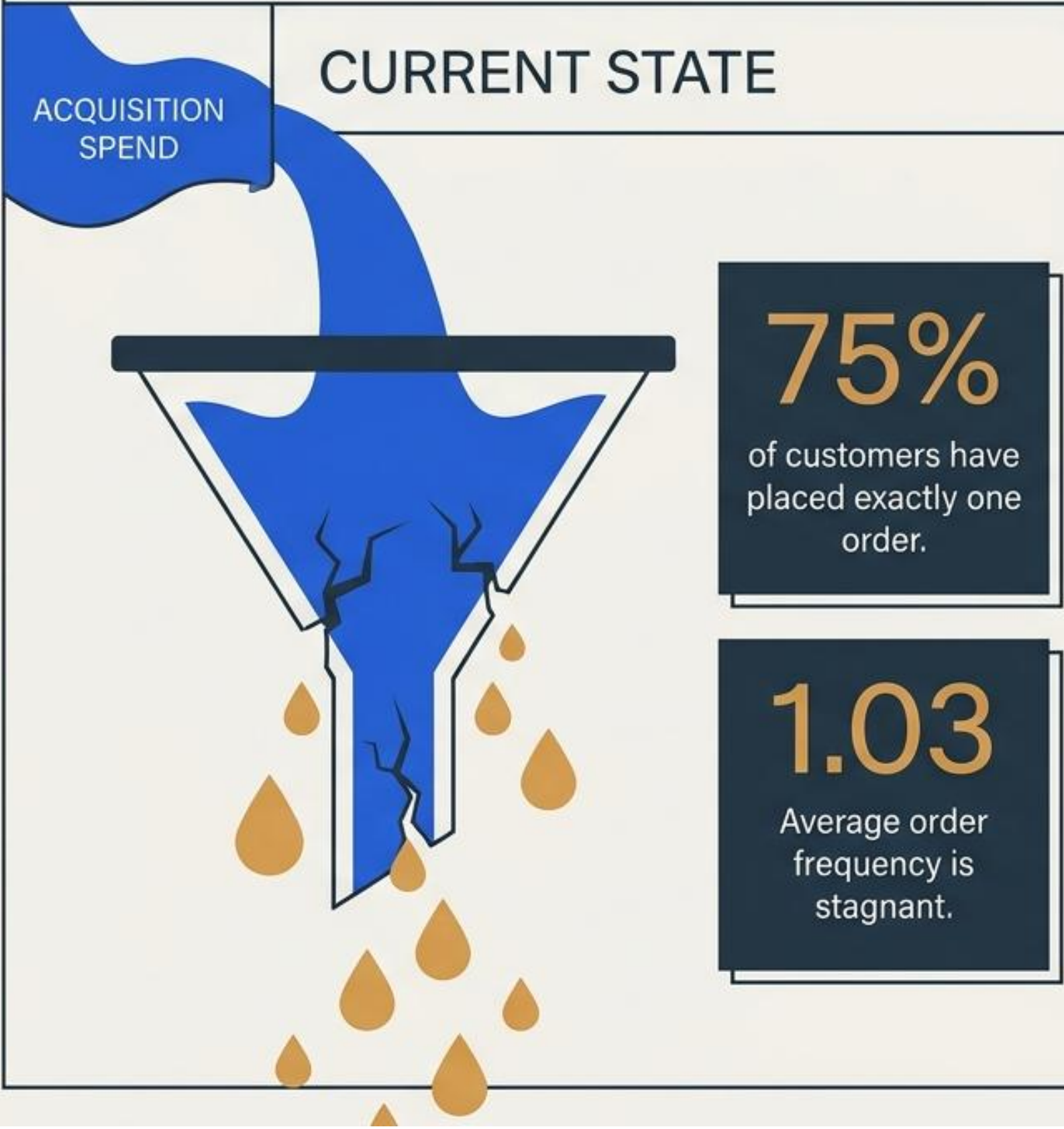


The Payoff

Pinpointed **R\$3.69M** in at-risk revenue and isolated the **6.5%** of 'Champion' customers driving **12.1%** of sales for targeted retention campaigns.

THE LEAKY BUCKET OF SINGLE-PURCHASE CUSTOMERS

CURRENT STATE



THE INTELLIGENCE GAP

CORE EXECUTIVE QUESTIONS

1.	Who are our most valuable customers?
2.	Which segments are quietly churning?
3.	Where should we allocate marketing budget for maximum ROI?

The Raw Material Scaling the Olist Operation

8

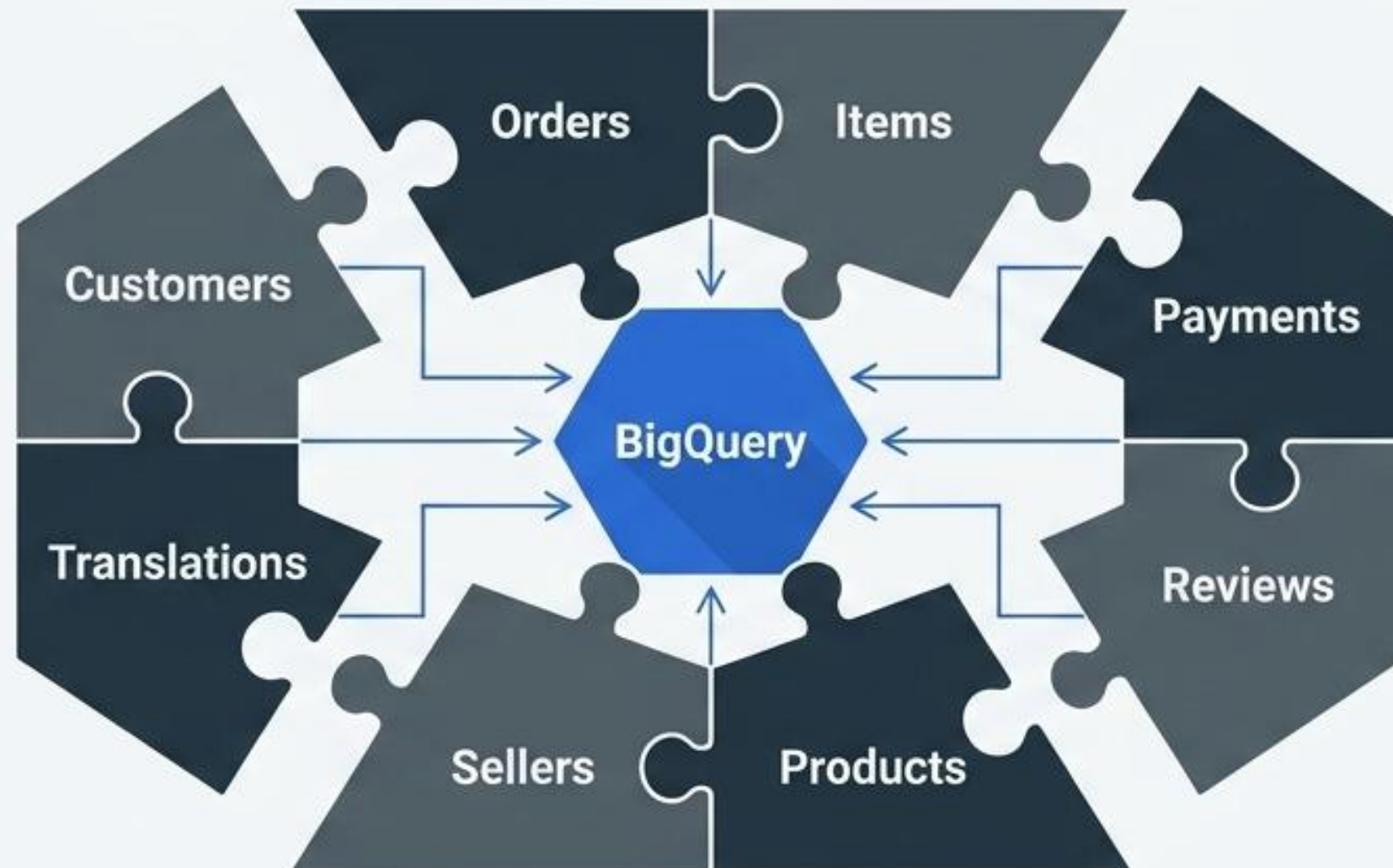
Interlocking Tables

~550,785

Total Records

3 Years

(Sep 2016 - Oct 2018)



Data Realities



Timestamps loaded as flat text strings.



Multiple payment installments causing potential revenue fan-out.

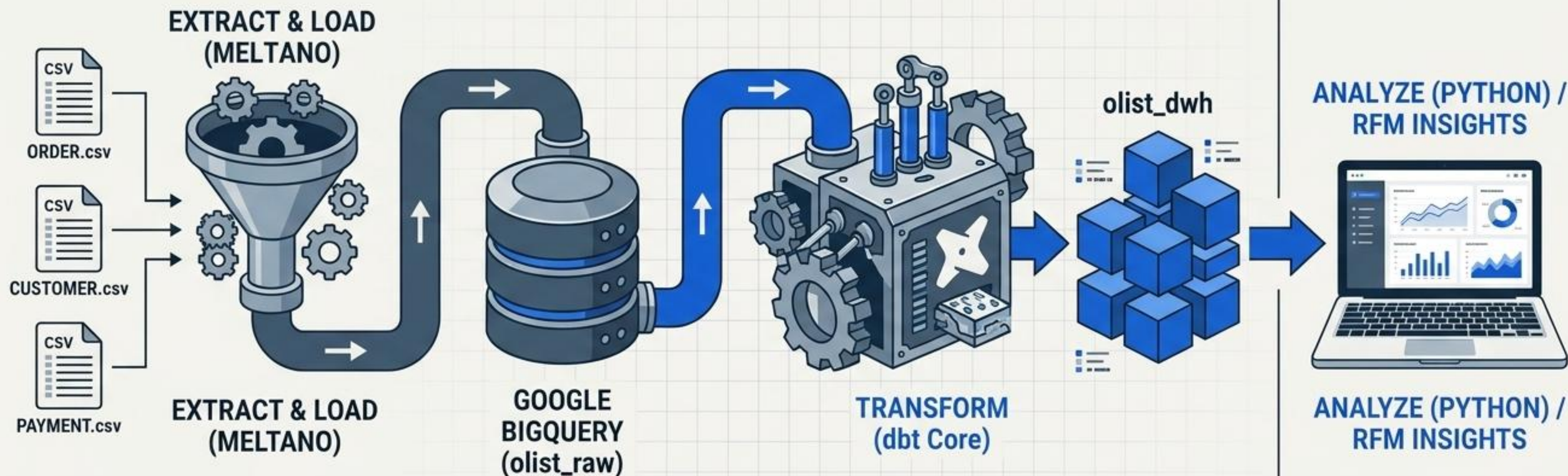


Composite primary keys requiring careful join logic.



A 1.6% baseline of missing product categorizations.

THE ANALYTICS REFINERY END-TO-END ARCHITECTURE



KEY ARCHITECTURAL BENEFITS

DECOUPLED COMPUTE & STORAGE

Tools operate independently without resource starvation.

IDEMPOTENCY

Pipeline can re-run endlessly without duplicating data.

ACCESSIBILITY

Analysts query clean marts, completely skipping the raw data chaos.

Securing the Raw Layer with Automated Ingestion

Custom Python Scripts ❌

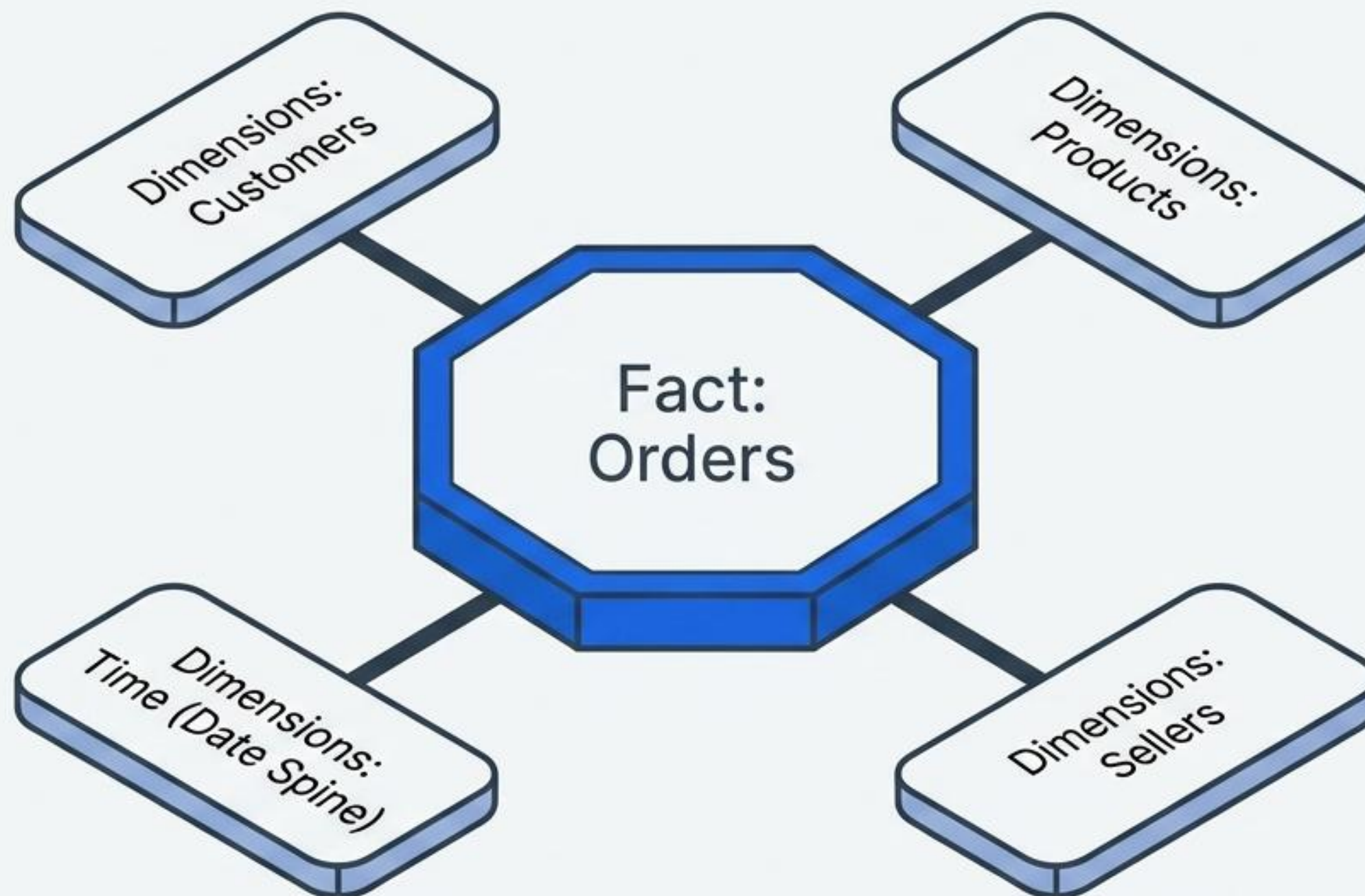
- Requires manual schema maintenance.
- Fragile error handling.
- High maintenance overhead for future engineers.

Meltano ELT ✅

Plugins Used: tap-spreadsheets-anywhere to target-bigquery

- ✅ Handles schema inference, batch uploading, and incremental states out-of-the-box.
- ✅ Establishes a stable, untouched raw layer in BigQuery.
- ✅ Preserves an absolute audit trail before any business logic is applied.

The Data Warehouse Blueprint Reorganizes Chaos



Strategic Rationale

Query Efficiency



Eliminates complex multi-table joins for analysts.

Data Integrity



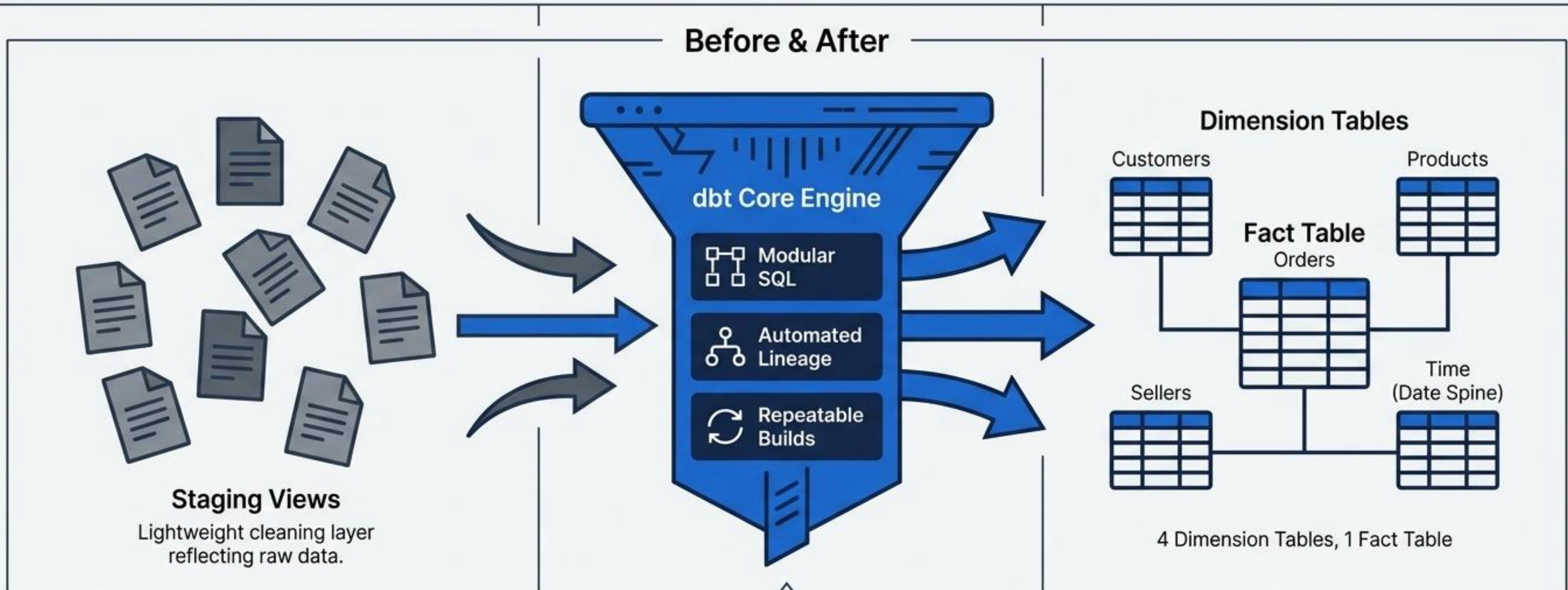
Surrogate and primary keys enforce a single source of truth.

Scalability



As Olist grows, the central fact table scales vertically while dimensions seamlessly absorb attribute updates.

Forging the Star Schema via dbt Transformations



Key Design Decision: RFM Decoupling

RFM segmentation was intentionally kept out of the warehouse marts. It remains an analytical output, allowing analysts to dynamically adjust scoring thresholds in Python without rebuilding upstream engineering.

Standardizing the Foundation Before Analytics Begin



Text Standardization

Trimmed messy IDs,
lowercased order statuses,
and mapped Portuguese
category names to English
presentation equivalents.



Type Casting

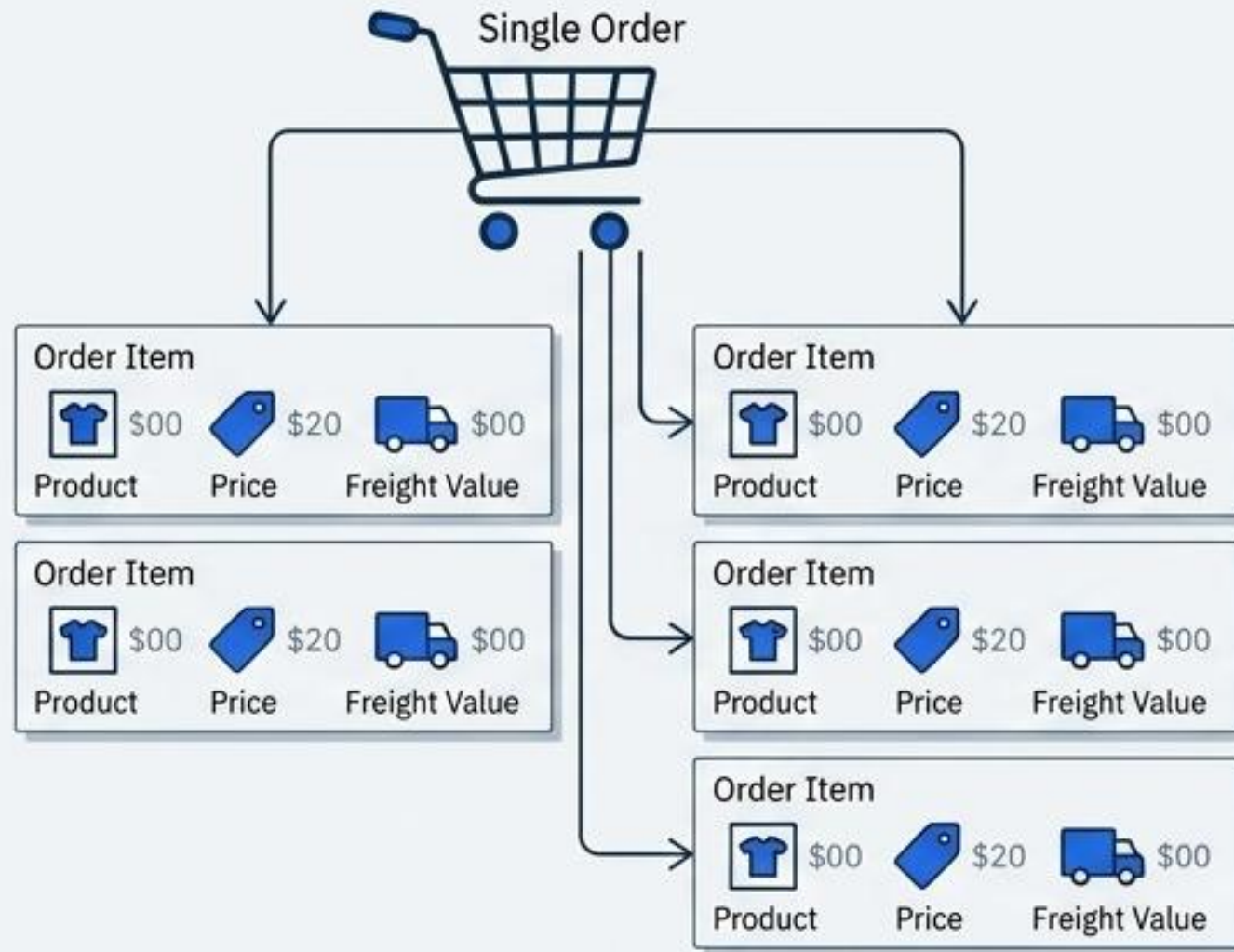
Forcibly converted raw string
timestamps into robust
Date/Time formats, enabling
accurate time-series
analysis.



Numeric Validation

Cast financial strings
(prices, freight, payments)
into strict numeric types,
systematically filtering out
invalid states.

Engineering Features for Accurate Revenue Attribution



Total Revenue = Price + Freight Value

The Grain Decision

The central Fact table was built at the Order-Item Grain, rather than the Order level.



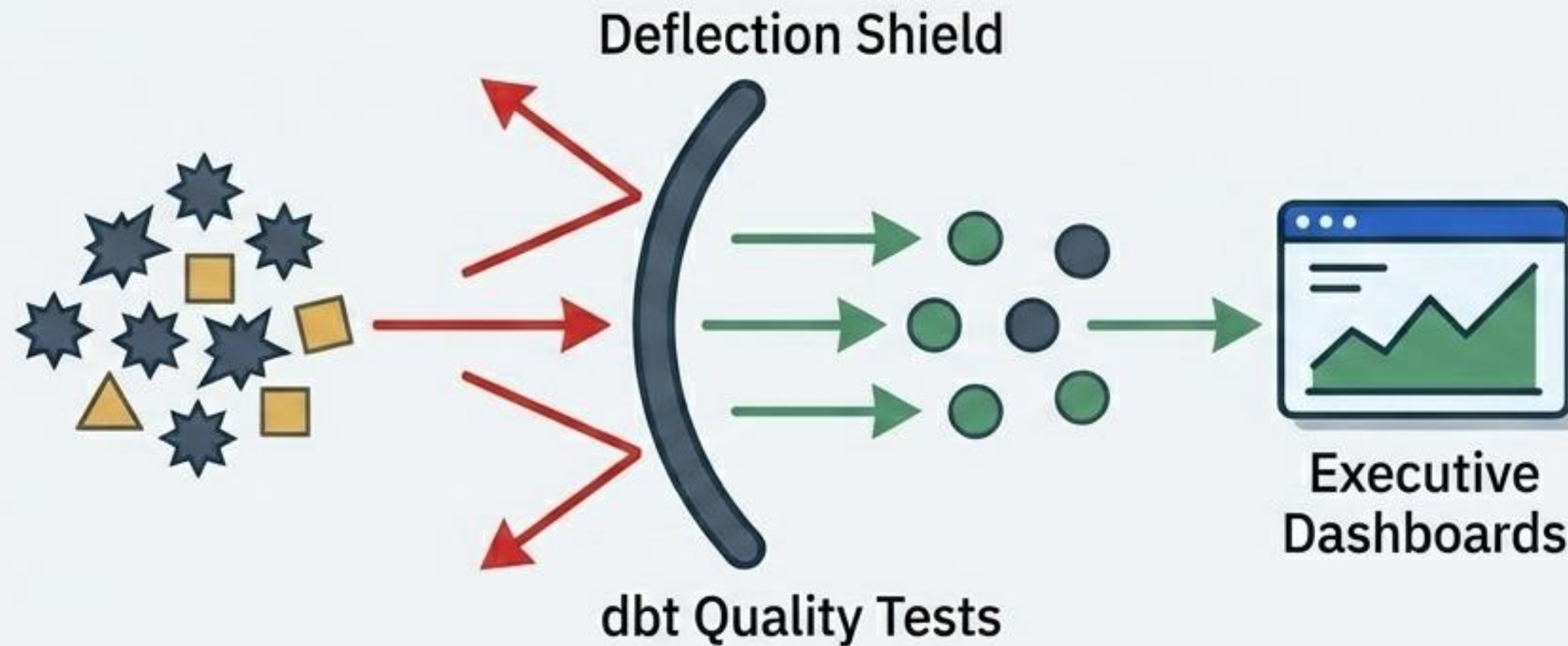
Why It Matters

Because Olist allows multiple distinct products within a single order, aggregating at the order level would erase product-level sales detail.

Engineering the warehouse at the item grain allows analysts to isolate exactly which products and sellers drive the highest freight costs or profit margins.



Automated Guardrails Protecting Business Intelligence



The Quality Philosophy

Data quality testing must happen inside the same workflow as the transformations.

Tests act as an automated shield, preventing bad insights from driving bad business decisions.

Why Native BigQuery Compute?

- Zero data movement.
- No separate execution environments required.
- Catches silent defects before they contaminate downstream Python analysis.

The Dual-Layer Testing Framework

Built-in Generic Tests (38 Tests)

Where

Defined in schema.yml



What It Validates

Absolute relationships. Checks for **Nulls**, **uniqueness**, and **Referential Integrity** (ensuring Foreign Keys map perfectly to Primary Keys).



Why

Irreplaceable for **guaranteeing the Star Schema's structural integrity**.



dbt-expectations (30 Tests)

Where

Defined in schema.yml
via **external package**



What It Validates

Business logic and shapes. Validates **value ranges**, **regex patterns**, **data types**, **statistical bounds**, and **exact row counts**.

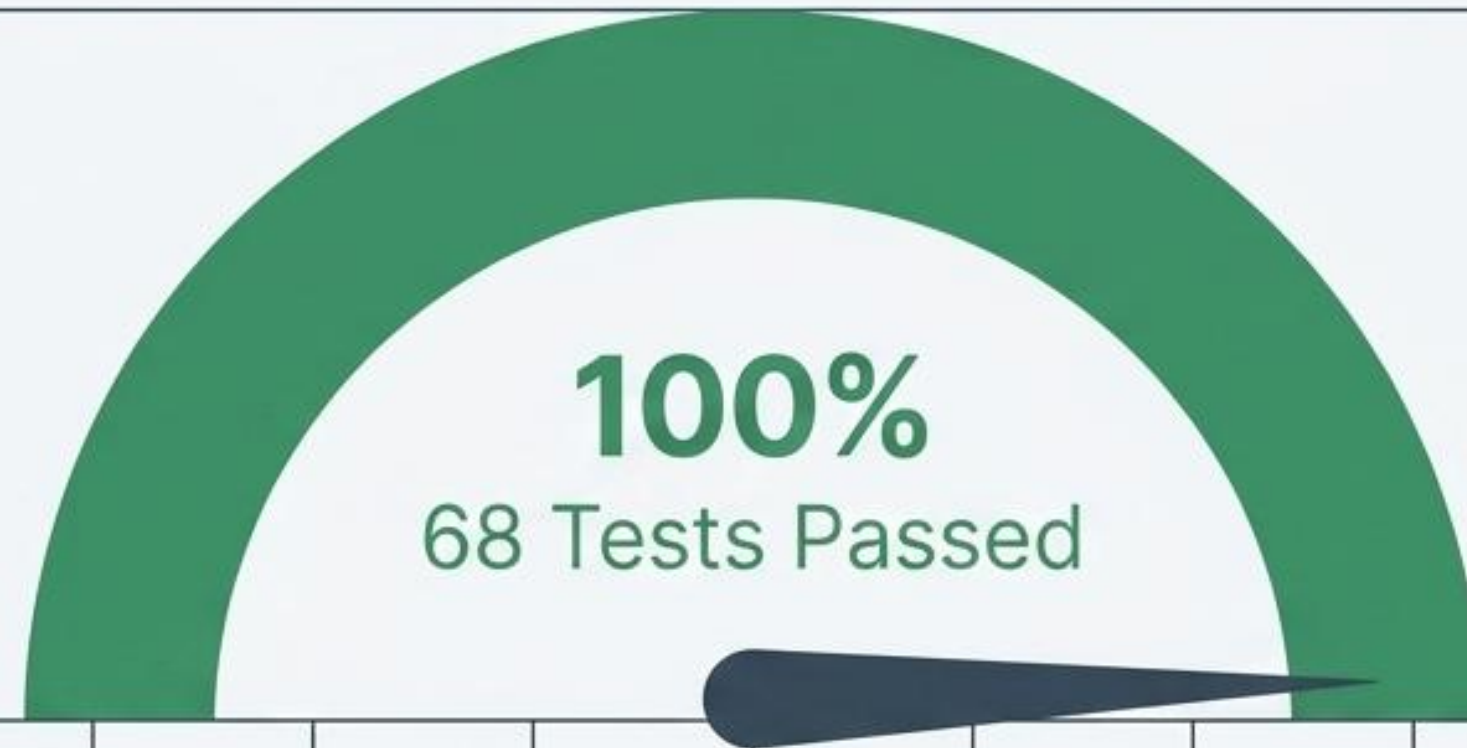


Why

Critical for **catching subtle data drift** (e.g., a sudden drop in average order value) that simple row-level checks miss.



The 68-Point Pre-Flight Inspection



Inspection Example



Financial Sanity

```
expect_column_values_to_be_between:  
[0, 100000]
```

Ensures payment values never drop below zero or spike to absurd system errors.

Inspection Example



Format Integrity

```
expect_column_value_lengths_to_equal:  
2
```

Guarantees customer state codes remain valid 2-letter Brazilian state identifiers.

Inspection Example

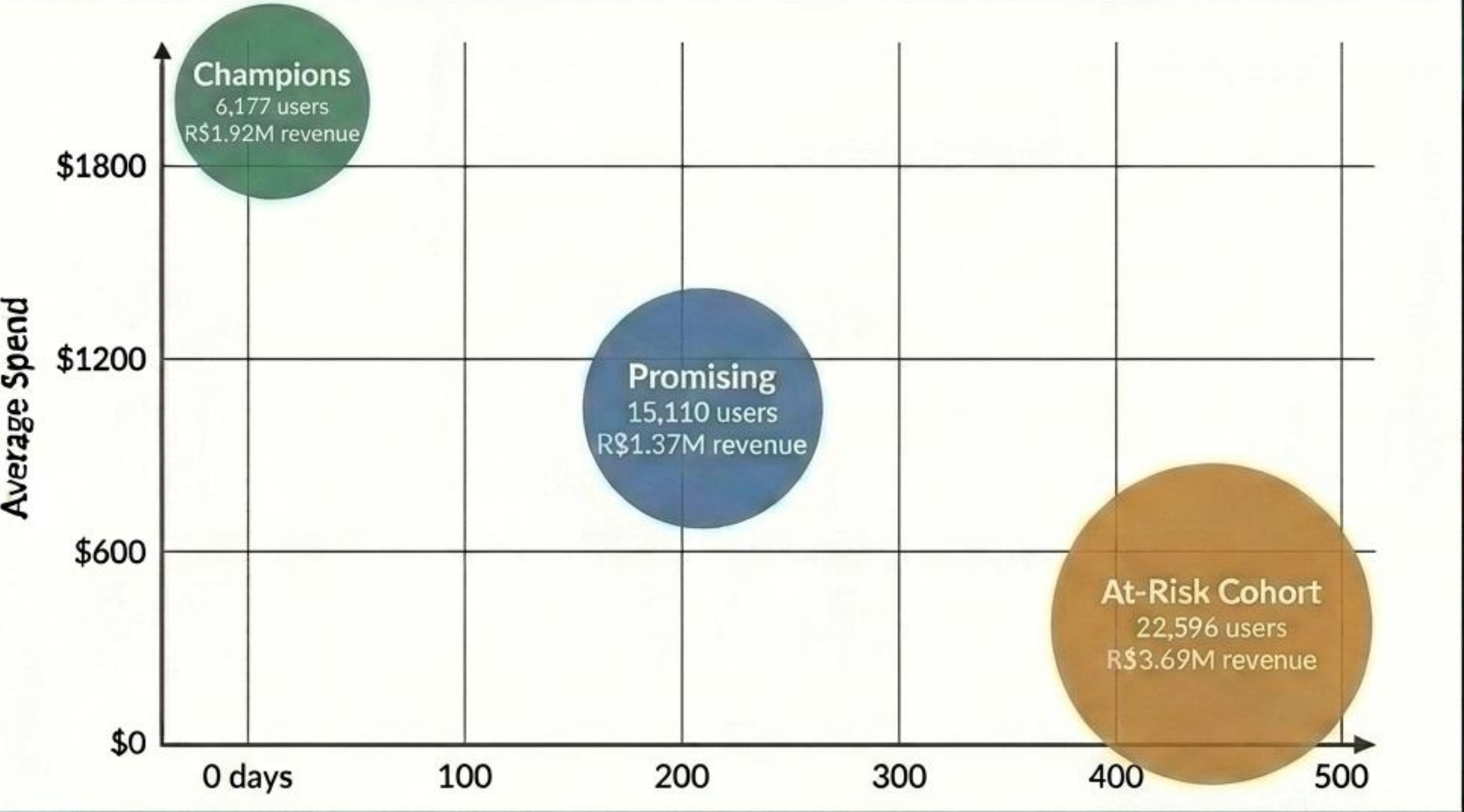


Catastrophic Loss Prevention

```
expect_table_row_count_to_be_between:  
[100000, 130000]
```

Acts as a smoke test against silent pipeline failures dropping massive chunks of Fact table data.

Decoding the Customer Base: The RFM Value Matrix



The Champions

- Only 6.5% of users, but driving 12.1% of revenue (R\$1.92M). High frequency, recent purchases, high spend.

The At-Risk Cohort

- 23.7% of users holding R\$3.69M in historical value.
- Have not purchased in over 400 days.



Key Insight: Bed, Bath & Table sells the highest volume of units, but Health & Beauty drives the highest total revenue (R\$1.44M)—proving retention must optimize for price point, not just unit volume.

Activating the Data: Three Strategic Priorities

Step 1: WIN BACK

 Metric



Target: At-Risk (**22,596** dormant customers)

 Metric



Action: Deploy highly targeted discount campaigns. Recovering even 10% of this segment protects **R\$369K** in historical revenue exposure.



 Metric

Step 2: PROTECT

 Metric



Target: Champions (**6,177** elite customers)



Action: Implement loyalty perks and priority service. Do not lose these outsized revenue drivers (averaging **R\$311** spend) to competitors.



 Metric

Step 3: CONVERT

 Metric



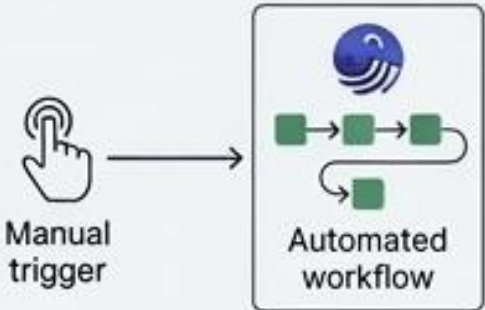
Target: Promising (**15,110** recent, single-purchase users)



Action: Launch automated second-purchase follow-up sequences to transition them into permanent 'Loyal Customer' status.

 Metric

Risk Mitigation and the Future Roadmap

Risks	Future Roadmap
The current ELT execution is highly reliable but requires manual triggering. Relying on manual execution delays time-to-insight and introduces operational risk as data volumes scale.	Automated Orchestration: Deploying Dagster to move the pipeline from manual triggers to event-driven, automated scheduling.  Geospatial Enrichment: Integrating reserved geolocation datasets to map regional delivery bottlenecks directly against customer churn rates. 

Executive Q&A Synthesis

Business Metrics

R\$15.8M

Total Historical Revenue Analyzed

R\$3.69M (23%)

Revenue At-Risk from dormant buyers

Technical Scale

550,785

Raw Records Ingested

12

Core Dimensional Models Deployed

Recommended Actions

1. Target At-Risk with win-back.
2. Insulate Champions with VIP perks.
3. Drive second purchases for Promising segment.

Governance

68 automated tests ensuring zero data anomalies.



Next Step: Dagster Orchestration.